

IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Visual impairment results in over 2.2 billion people worldwide with a considerable reduction in their independence and a 30-50% higher chance of accidents such as falling within their homes. Current IoT-based assistive technologies are mostly based on GPS, which cannot be used indoors, or on continuous connectivity to the cloud, which results in critical failures when the signal strength is low. Additionally, current guardian apps are mostly reactive, providing raw location information without context and are unable to identify “silent” emergencies such as medical emergencies without a fall occurring.

The current research aims to develop an Intelligent Security and Remote Guardian System that overcomes these limitations. The novelty of this system is its Context Aware Risk Assessment and Multimodal Emergency Verification. This will be achieved by incorporating real-time hazard detection with an AI-powered spectacle with YOLOv8-Tiny and a hybrid positioning system. This will allow for a real-time “Safety Score” along with contextual alerts such as “User near stairs.” Additionally, this research will incorporate a new Inactivity Monitoring protocol with IMU sensor fusion to detect unconsciousness without a fall.

The solution utilizes Flutter for a mobile user interface, Firebase for real-time data synchronization, and Raspberry Pi for edge computing to enable offline support. The expected outcome is a high-accuracy distress detection system with a Voice-Activated SOS feature to enable emergency alert in incapacitated conditions.

Social Impact: The system will improve self-reliance and mental security among the visually impaired. **Business Impact:** The system will capitalize on the growing multi-billion-dollar assistive technology industry. This research will add a strategic roadmap to improve reliability and situational awareness of guardian-assisted navigation in underserved indoor environments.

Keywords: Visually Impaired, Indoor Navigation, Remote Guardian System, Fall Detection, Inactivity Monitoring, Geo-fencing, IoT Safety, Multi-modal Sensors, Flutter.

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In addition, I would like to express my gratitude to my fellow group members for their cooperation and collaborative efforts in integrating the different components of the IoT-Based Spectacle system. The technical contributions of my fellow group members were instrumental in developing an integrated system across the hardware, AI, and mobile layers of the IoT-Based Spectacle system.

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LIST OF ABBREVIATIONS

Abbreviation	Description
API	Application Programming Interface
AI	Artificial Intelligence
FCM	Firebase Cloud Messaging
GPS	Global Positioning System
IDE	Integrated Development Environment
IMU	Inertial Measurement Unit
IoT	Internet of Things
IP	Intellectual Property
JSON	JavaScript Object Notation
LiDAR	Light Detection and Ranging
YOLO	You Only Look Once
MQTT	Message Queuing Telemetry Transport
WBS	Work Breakdown Structure
UUID	Universally Unique Identifier
NLP	Natural Language Processing

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1. INTRODUCTION

1.1. Background and Context

This is particularly significant in the socio-economic arena, with the World Health Organization (WHO) stating that more than 2.2 billion people suffer from some degree of vision loss. In this figure, 43 million are completely blind, which makes it very difficult for them to move around in their daily places of work or study, including homes, offices, and schools. In this context, the problem affects people like "Alex," who is blind and works in an office in Sri Lanka. The seriousness of this problem can also be seen in the fact that visually impaired and blind (VIB) people are 30% to 50% more likely to fall than sighted people, which often results in serious injuries. In addition to this, the socio-economic impact cannot be taken lightly, with losses estimated at 410 billion dollars annually.

Assistive technology has progressed from mechanical devices such as the white cane to electronic travel aids that incorporate ultrasonic technology. There are, however, some limitations that have been observed in current assistive technology, where traditional technology cannot sense multiple hazards, and current sophisticated technology relies on internet connectivity, which may not function in indoor environments with poor connectivity. Moreover, current guardian applications are passive, providing raw location information without any context to help the guardian understand the presence of hazards such as an upcoming staircase.

To bridge the gap, the present research aims to develop an IoT-based Smart Spectacle System, which can function as an umbrella for the visually impaired. The system is designed to be wearable, cost-effective, and utilizes edge computing to ensure effective offline operation without relying on cloud connectivity. The system includes several specialized components, such as an AI Hazard Detection Engine, powered by YOLOv8-Tiny, and a Hybrid Indoor Positioning System, in addition to the Smart Hardware Module. The heart of the system is the Intelligence Security and Remote Guardian System, designed to function as an intermediary between the visually impaired person and their caregiver, using multi-layered safety features such as inactivity detection, fall detection, and SOS calls using voice assistance.

The current research is strategically related to the Autonomous Intelligent Machines and Systems (AIMS) research group of SLIIT, as the group deals with smart autonomous systems. In addition, the proposed project is related to the following United Nations Sustainable Development Goals (SDGs): Goal 3, Good Health, and Well-being, by reducing the risk of injuries; Goal 10, Reduced Inequalities, by promoting independence; and Goal 11, Sustainable Cities and Communities, by enhancing mobility.

1.2. Literature Survey

The progression of assistive technology for visually impaired and blind (VIB) people has progressed from mechanical aids to sophisticated electronic travel aids (ETAs) that incorporate the Internet of Things (IoT) and Artificial Intelligence (AI) technologies. Recent trends in VIB research highlight the need for a holistic "Smart Spectacle" ecosystem, which incorporates real-time obstacle avoidance, indoor positioning, and remote guardian tracking. These advanced assistive technologies aim to overcome the limitations of the traditional white cane by incorporating sensor fusion techniques to detect complex obstacles and changes in elevation. These include LiDAR, camera, and Inertial Measurement Unit (IMU) sensors.

1.3. Analysis of Existing Research and Products

Scholarly research from 2024 to 2026 emphasizes the use of edge-computing to reduce the latency of hazard detection. For instance, studies on YOLO (You Only Look Once) models, specifically YOLOv8-tiny, have demonstrated high precision in identifying indoor obstacles like furniture and stairs while running on low-power devices like Raspberry Pi. Furthermore, existing products such as the **OrCam MyEye** utilize advanced computer vision for object recognition and reading assistance. Similarly, the **WeWALK smart cane** integrates ultrasonic sensors with smartphone navigation to provide haptic feedback to the user. Research in sensor-based fall detection highlights a shift toward multiple-sensor systems, which are proven more efficient for indoor rescue operations than single-accelerometer designs.

1.4. Challenges in Indoor Navigation and Mobile Support

Traditional navigation systems heavily rely on the Global Positioning System (GPS); however, its effectiveness is drastically reduced in indoor environments due to signal attenuation caused by physical structures. Research highlights that for VIB individuals, navigating unfamiliar indoor spaces such as hospitals or offices poses a significant risk of collisions and falls. Existing smartphone navigation apps often lack rigorous hazard analysis and risk assessment, failing to provide proactive safety measures within a home or office environment. Furthermore, many current apps provide raw location data but fail

to integrate environmental context, leaving the guardian unaware of specific hazards like stairs or dynamic obstacles nearby.

1.5. Evolution of Fall Detection and Inactivity Monitoring

Accidental falls are a leading cause of injury and loss of autonomy among the elderly and VIB population. Literature classifies fall detection systems into two main categories: single-sensor and multiple-sensor-based systems. While single-sensor systems (specifically those using accelerometers) are preferred for wearable devices due to low power consumption, research indicates that multiple-sensor-based systems are more efficient and accurate for comprehensive indoor monitoring.

A critical gap identified in the literature is the focus on "impact-based" fall detection while neglecting "inactivity monitoring." Many systems can detect a sudden fall, but they fail to identify medical emergencies where a user remains stationary or unconscious without a high-impact event. The integration of IMU (Inertial Measurement Unit) data for both fall detection and inactivity monitoring is essential for a robust safety protocol.

1.6. Voice-Activated SOS and Emergency Communication

The convergence of AI and wearable technology has introduced voice-activated distress signals, which are transformative for personal safety. Advanced voice recognition AI can now distinguish distress signals from regular speech, allowing users to trigger an SOS even when physically incapacitated or unable to reach a physical button. However, the effectiveness of these systems is often limited by their reliance on cloud connectivity. As noted in recent studies, assistive devices like OrCam MyEye face challenges in offline reliability, necessitating edge-computing solutions that process safety alerts locally on the device.

1.7. Critical Evaluation and Limitations

Despite these advancements, a critical evaluation of existing systems reveals several recurring limitations:

- **Infrastructure and Connectivity Dependency:** Many indoor navigation research approaches rely on pre-installed BLE beacons or Wi-Fi tags, which are impractical for large-scale deployment in developing regions.
- **Cloud Reliance:** Commercial products like OrCam often require cloud connectivity for intensive processing, leading to "safety blackouts" in low-signal indoor areas.

- **Reactive Nature of Monitoring:** Most guardian services and SOS apps are purely reactive, providing raw location data after an incident has occurred, but lacking the "Contextual Intelligence" (e.g., "User near stairs") required for proactive safety.
- **The Inactivity Gap:** Existing fall detection literature focuses heavily on high-impact collisions, frequently neglecting "Inactivity Monitoring" for medical emergencies where the user remains stationary without a physical fall.
- **Linguistic Barriers:** A significant portion of existing Voice-SOS research is optimized for English, creating a gap in localized support for regions like Sri Lanka.

1.8. Research Gap

The development of indoor assistive technologies has primarily focused on isolated functional units, such as basic obstacle detection or GPS-based tracking, rather than a unified, context-aware safety ecosystem. While current "Smart Spectacle" research introduces advanced vision-based hazard detection, there remains a critical disconnection between the user's wearable hardware and the guardian's remote interface. Most existing systems operate as passive data-loggers that lack the intelligence to synthesize raw sensor data into actionable safety insights, leaving a significant void in proactive emergency management.

The insufficiency of current approaches and the specific gaps this research addresses are detailed below:

- **Missing Contextual Intelligence and Proactive Monitoring:** Current remote monitoring applications typically provide only raw location coordinates (Latitude/Longitude), which are insufficient in complex indoor settings where a guardian cannot see the immediate hazards surrounding the user. There is a complete absence of systems that provide "Contextual Awareness," such as notifying a guardian that a user is specifically near a staircase or a high-risk elevation change.
- **The "Inactivity" Detection Void:** Existing safety wearables are heavily biased toward "impact-based" fall detection. Current research fails to provide reliable protocols for "silent" medical emergencies—situations where a user remains stationary or loses consciousness without a high-impact fall. This gap in inactivity monitoring often leads to delayed rescue responses in unsupervised environments.

- **Infrastructure and Language Barriers:** Current high-end solutions often require pre-installed indoor infrastructure (like BLE beacons) or high-speed cloud connectivity, which are neither practical nor affordable in developing regions like Sri Lanka. Furthermore, most voice-activated SOS research is optimized for English, creating a linguistic barrier for local users who require localized auditory triggers and guardian alerts.
- **Justification of Novelty and Creativity:** The novelty of this research lies in its **Multi-Layered Safety Protocol** that fuses IMU-based inactivity monitoring with real-time AI hazard metadata. By shifting the computation to an **Offline-Capable Edge Hub** (Raspberry Pi), the system creatively bypasses the "cloud-dependency" failure point found in commercial products. This integration ensures that the mobile app acts as an intelligent bridge, providing guardians with a "Safety Score" and environmental context that is currently non-existent in the market.

2. OBJECTIVES

2.1. Main Objective

The primary objective of this research component is to design, develop, and evaluate robust. Intelligence Security and Remote Guardian System integrated into a cross-platform mobile application. This system acts as a centralized safety hub that processes multi-modal sensor data from a smart spectacle to provide guardians with real-time, context-aware monitoring and automated emergency response protocols for visually impaired users indoors environments.

2.2. Special Objectives

To achieve the main objective, the following specific objectives have been identified:

- **Design and Implementation of Multi-Layered Safety Algorithms:** To develop and integrate sophisticated algorithms for Fall Detection and Inactivity Monitoring using fused data from the spectacles' Inertial Measurement Unit (IMU) to distinguish between accidental impacts and medical emergencies where the user remains motionless without a fall.
- **Development of a Context-Aware Guardian Dashboard:** To create a real-time monitoring interface for guardians (using Flutter/React Native) that visualizes the user's precise indoor location alongside environmental hazards (e.g., "User near stairs") identified by the AI vision module. **Integration of a Hybrid SOS Communication Workflow:** To implement a redundant emergency triggering mechanism that combines a Voice-Activated SOS (utilizing natural language processing) and a physical panic button, ensuring a hands-free distress signal is transmitted even when the user is physically incapacitated.
- **Implementation of Adaptive Geo-fencing and Safe-Path Tracking:** To develop a geo-fencing logic that allows guardians to define "Safe Zones" and receive immediate alerts if the user deviates from learned indoor paths or enters high-risk areas.
- **Establishment of a Real-Time Data Synchronization Hub:** To integrate the mobile application with a cloud-based database (Firebase) to ensure near-zero latency in transmitting location logs, battery status, and emergency alerts from the edge-processing unit to the remote guardian.
- **Validation through User Acceptance Testing (UAT):** To evaluate the system's reliability, usability, and response time in various indoor scenarios to ensure it meets the safety requirements of visually impaired individuals and their caregivers.

3. Methodology

The proposed methodology for the **Intelligence Security and Remote Guardian System** follows the **Design Science Research (DSR)** methodology. This approach is chosen as the project focus is on creating and evaluating a new "artifact"—the Intelligence Security and Remote Guardian System—to solve a specific real-world problem (indoor safety for VIB users). The process involves iterative cycles of designing safety protocols, prototyping the mobile hub, and validating its effectiveness through rigorous testing.

3.1. System Architecture

The architecture follows a three-tier model: **Edge Tier, Cloud Tier, and Mobile Tier.**

- **Edge Tier (Hardware Integration):** The Raspberry Pi processing unit collects raw data from the IMU, LiDAR, and Camera modules.
- **Cloud Tier (Data Management):** Firebase is utilized as a real-time database and authentication service to synchronize alerts and location logs with minimal latency.
- **Mobile Tier (Guardian Interface):** A cross-platform application (developed using Flutter/React Native) serves as the primary dashboard for the guardian to receive real-time updates and trigger emergency responses.

3.2. Requirement Analysis

To ensure the system meets the safety needs of VIB users, a comprehensive requirement analysis was conducted:

- **Functional Requirements:** The app must support real-time location tracking, fall/inactivity detection alerts, voice-activated SOS triggers, and geo-fencing configuration.
- **Non-Functional Requirements:** High availability, low-latency alert transmission (under 2 seconds), and data privacy (anonymized logs) are critical for user safety.

3.3. Design and Implementation of Safety Protocols

This section details the logic behind the multi-layered security features:

- **Fall and Inactivity Detection Logic:** A threshold-based algorithm is implemented using fused accelerometer and gyroscope data from the IMU sensor. If no movement is detected for a predefined period after an impact (or without an impact), an "Inactivity Alert" is escalated to the guardian.
- **Context-Aware Alerting:** The mobile app processes object detection strings (e.g., "Stairs", "Obstacle") received from the AI module (YOLOv8-Tiny) to display environmental context on the dashboard.
- **Voice-Activated SOS Workflow:** Utilizing natural language processing (NLP), the app listens for specific distress keywords (e.g., "Help", "Emergency") to bypass manual interaction and initiate the rescue protocol.

3.4. Development Lifecycle (Agile Methodology)

The project follows the **Agile Development Model**, allowing for iterative improvements based on testing:

1. **Sprints:** Development is divided into 2-week sprints, focusing on individual features like Firebase integration, UI design, and sensor calibration.
2. **Prototyping:** Incremental prototypes are tested to verify the communication link between the Raspberry Pi and the mobile application.

3.5. Tools and Platforms with Justification

- **Flutter (Mobile Framework):** Chosen for cross-platform compatibility (iOS/Android) and its high-performance rendering, which is critical for real-time dashboard updates.
- **Firebase (Backend & Real-time Database):** Selected for its NoSQL Realtime Database capabilities, ensuring that emergency alerts are synchronized between the spectacles and the guardian app with sub-second latency.
- **Python (Edge Logic):** Used on the Raspberry Pi for efficient data pre-processing and handling the communication bridge (Web Sockets/MQTT) to the mobile app.

- **VS Code & Android Studio:** Utilized as the primary Integrated Development Environments (IDEs) for coding and debugging.

3.6. Data Collection and Ethical Considerations

- **Data Collection:** Initial data consists of IMU sensor readings (accelerometer/gyroscope) recorded during simulated falls and stationary periods to calibrate the inactivity threshold.
- **Ethical Considerations:** All user data, including indoor location logs and emergency voice recordings, are handled with strict confidentiality.
 - **Informed Consent:** Participants in the UAT phase will provide explicit consent.
 - **Data Privacy:** Firebase authentication and encrypted data transmission (TLS 1.3) are used to ensure only authorized guardians can access the VIB user's data.

3.7. Testing, Validation Strategy and Performance Metrics

To ensure reliability, the system undergoes rigorous testing:

- **Unit Testing:** Testing individual modules such as the SOS trigger and the geo-fencing logic.
- **Integration Testing:** Verifying the full pipeline from sensor input (spectacles) to output (mobile alert).
- **Performance Testing:** Measuring the battery consumption of the mobile device and the latency of emergency notifications.

The system will be validated using a **Quantitative Testing Strategy** based on the following metrics:

- **Detection Accuracy:** Measuring the Precision and Recall of the fall and inactivity algorithms (Target: >90%).
- **Latency Measurement:** Testing the time taken from a hardware-detected event to the guardian's mobile notification (Target: <2 seconds).
- **System Reliability:** Evaluating the app's performance in "Offline Mode" by checking the consistency of local SQLite logging when Wi-Fi is disconnected.

- **User Acceptance Testing (UAT):** Gathering feedback from 5-10 potential guardians to evaluate the usability of the Context-Aware Dashboard through a System Usability Scale (SUS) survey.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1. Overview of Proposed Solution

The proposed solution is an integrated IoT-based spectacle ecosystem that utilizes Edge Computing, AI-driven hazard detection, and a specialized Mobile Hub. The system is designed to provide visually impaired users with independent indoor navigation while maintaining a persistent safety link with a remote guardian.



Figure 1: Overview of the proposed system

The proposed system functions as an end-to-end safety ecosystem designed to bridge the gap between visually impaired users and their remote guardians. The integration is

structured to ensure that every environmental hazard detected by the wearable is translated into a meaningful insight for the caregiver.

- **Sensing and Perception:** The cycle begins at the smart spectacle tier, where multi-modal sensors (LiDAR, Camera, IMU) continuously capture environmental and motion data.
- **Edge Processing:** To ensure offline reliability, raw data is processed locally on a Raspberry Pi unit. This unit executes the YOLOv8-Tiny model for hazard detection and runs the primary fall/inactivity detection logic.
- **Cloud Synchronization:** Processed alerts and location metadata are pushed to the Firebase Real-time Database. This tier ensures that even if the guardian is miles away, they receive updates with sub-second latency.
- **Guardian Engagement:** The mobile application (Individual Component) serves as the final interface. It transforms raw data into a **Context-Aware Dashboard**, enabling guardians to visualize the user's surroundings and respond to voice-activated SOS signals or automated emergency alerts.

4.2. System Diagram of Components

The system architecture is categorized into three primary tiers:

1. **Edge Tier (Hardware):** Comprising the Smart Spectacle (Raspberry Pi, LiDAR, Camera, and IMU) which manages real-time environment sensing and local AI processing.
2. **Cloud Tier (Backend):** Utilizing Firebase for real-time data synchronization, user authentication, and persistent alert logging.
3. **Mobile Tier (Guardian Interface):** The Intelligence Security and Remote Guardian mobile application (Flutter) that visualizes data for the caregiver.

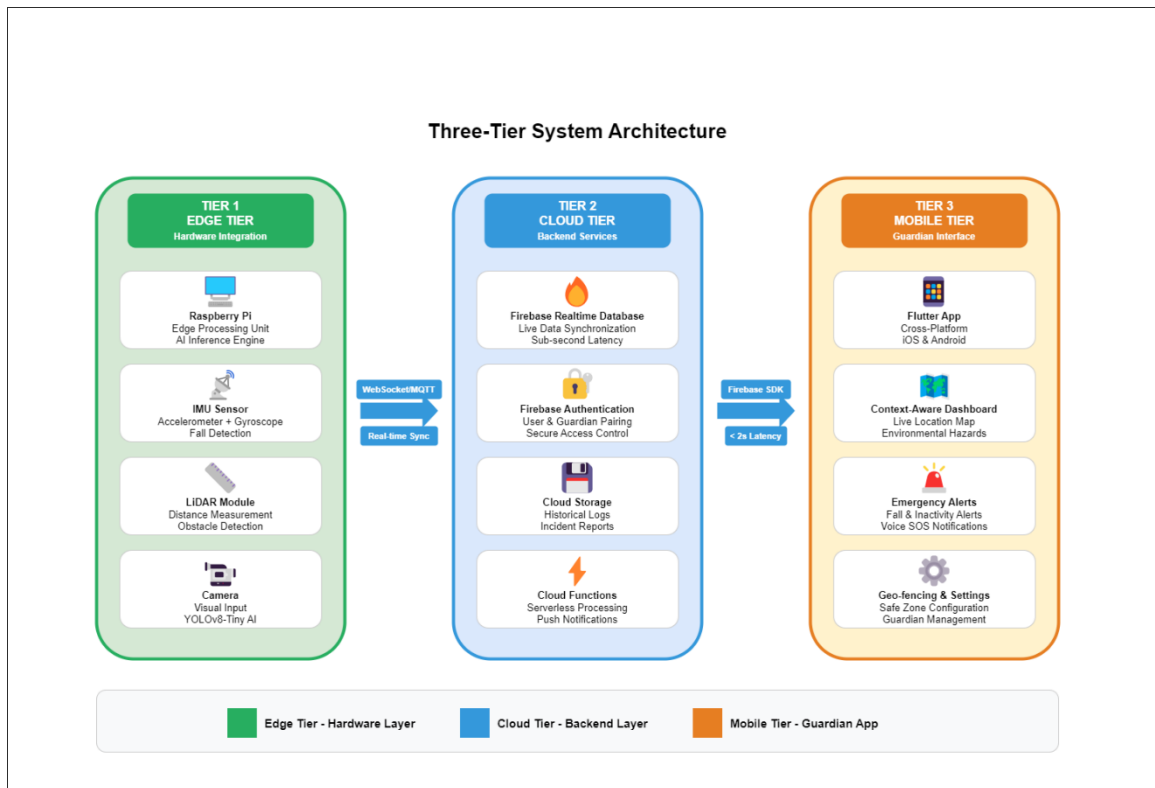


Figure 2: System Architecture

4.3. Individual Component Focus: Intelligence Security & Remote Guardian System

My individual component acts as the **central intelligence hub**. It is responsible for:

- **Sensor Fusion Logic:** Processing IMU data for fall and inactivity detection.
- **Verification Engine:** Using voice NLP to confirm emergencies.
- **Guardian Dashboard:** A high-level UI that displays the "Safety Score" and contextual hazards.

4.4. Component Interactions and Data Flow

1. **Sensing to Edge:** Raw sensor data (100Hz IMU) is sent to the Raspberry Pi.

2. **Edge to Cloud:** Processed hazard metadata and emergency triggers are pushed to Firebase via MQTT/WebSockets.
3. **Cloud to App:** The Guardian App observes Firebase snapshots to update the dashboard in under 2 seconds.
4. **Feedback Loop:** The guardian can send voice guidance or trigger haptic feedback back to the user via the app.

4.5. Scalability and Security Considerations

- **Scalability:** The use of Firebase allows the system to scale horizontally to support multiple user-guardian pairs. The modular architecture ensures that additional sensors (e.g., heart rate monitors) can be integrated without redesigning the core app.
- **Security:** Data is protected using TLS 1.3 encryption for all transmissions. Access control is managed through Firebase Auth, ensuring that only authorized guardians can view the VIB user's real-time location and safety history.

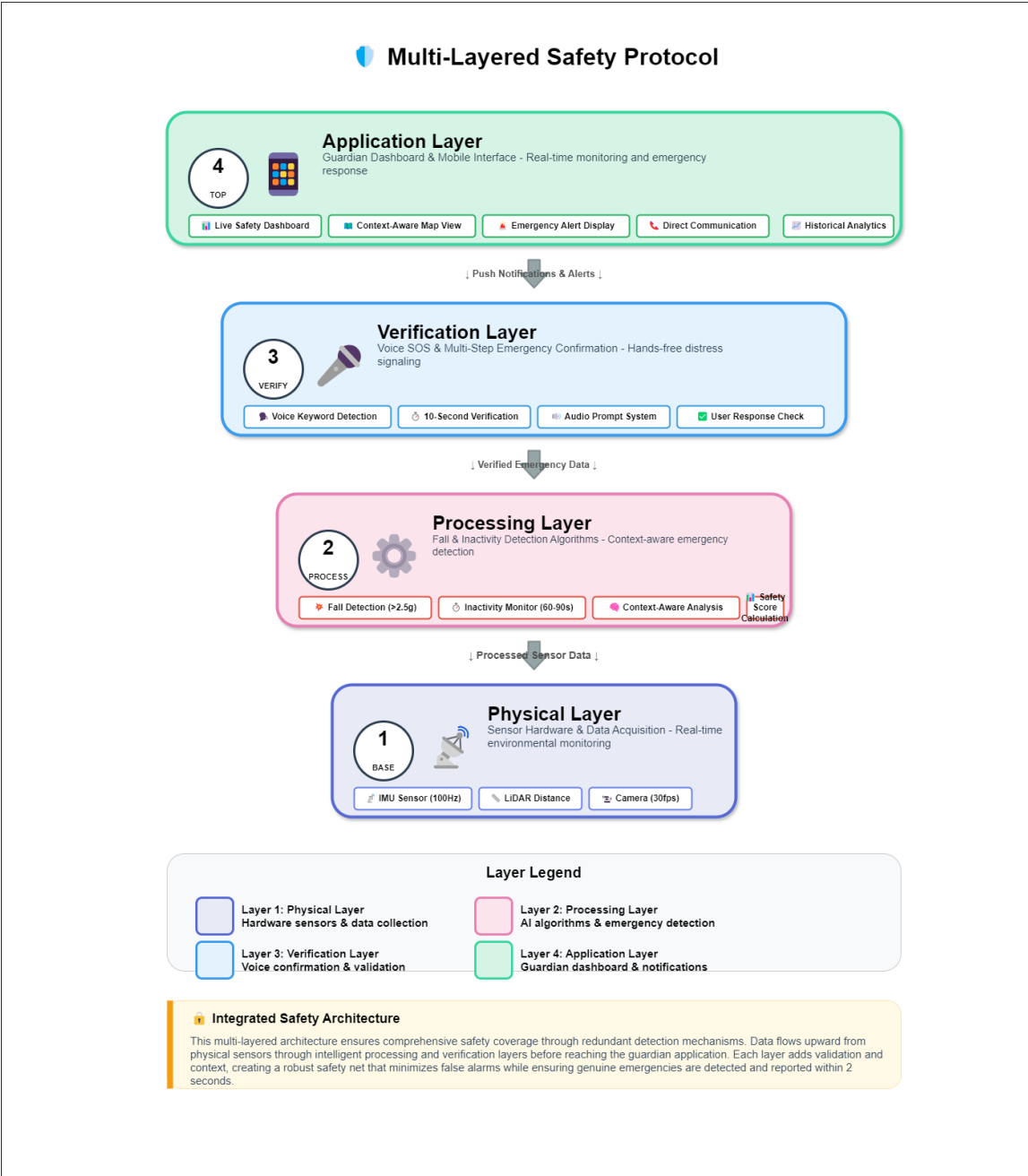


Figure 3: Safety protocol System.

5. User Requirements

The requirements for this component focus on the mobile application's ability to act as a safety hub between the wearable hardware and the human guardian.

5.1. Stakeholder Identification

This system involves multiple stakeholders who interact with the application either directly or indirectly:

- **Primary Stakeholder (Visually Impaired User):** The individual wearing the smart spectacles. They require a hands-free, reliable safety system that works without constant manual intervention.
- **Secondary Stakeholder (Remote Guardian/Caregiver):** The primary user of the mobile application. They need real-time, context-aware data to monitor and assist the VIB user.
- **Tertiary Stakeholders (Emergency Services):** Local authorities or medical responders who receive escalated SOS alerts and location data in critical situations.
- **Developers/Researchers:** Responsible for system maintenance, sensor calibration, and ensuring the reliability of the safety algorithms.

5.2. User Requirements

- **Guardian Accessibility:** The guardian must be able to monitor the user's real-time indoor location and safety status remotely via a mobile dashboard.
- **Immediate Notification:** The user (visually impaired individual) requires an automated system that alerts their designated guardian during emergencies without manual interaction.
- **Contextual Information:** The guardian requires environmental context (e.g., "User near stairs") to provide better verbal assistance via the app.

5.3. Functional Requirements

The functional requirements are categorized based on the core intelligence of the mobile component:

1. **FR-01: Real-time Hazard Visualization:** The system shall display real-time environmental hazards (e.g., "Stairs", "Obstacle") on the guardian dashboard using metadata from the AI vision module.
2. **FR-02: Fall and Inactivity Detection:** The system shall process 3-axis IMU data to detect sudden impacts or prolonged periods of user inactivity (>60 seconds) and trigger an automated verification sequence.
3. **FR-03: Voice-Activated SOS Trigger:** The system shall allow the user to initiate a distress signal using specific natural language keywords (e.g., "Help", "Emergency") recognized by the mobile app's NLP engine.
4. **FR-04: Adaptive Geo-fencing:** The system shall enable guardians to define digital "Safe Zones" on a map and generate immediate push notifications if the user exits these boundaries.
5. **FR-05: Contextual Risk Scoring:** The system shall calculate and display a real-time "Safety Score" for the guardian, derived from a weighted analysis of nearby hazards, battery levels, and connectivity status.
6. **FR-06: Remote Communication Bridge:** The app shall provide a direct audio/voice link between the guardian and the user once an emergency alert is triggered.
7. **FR-07: Persistent Data Logging:** The system shall synchronize all safety events, location logs, and sensor triggers with the Firebase Real-time Database for historical analysis.

5.4. Non-Functional Requirements

These requirements ensure the system's effectiveness, reliability, and ease of use:

- **Performance (Latency):** The system must ensure that the end-to-end latency from an emergency detection on the hardware to a notification on the guardian's app is less than 2 seconds.
- **Security & Privacy:**
 - All data transmissions between the Raspberry Pi, Firebase, and the Mobile App must be encrypted using TLS 1.3.
 - Access to the user's real-time location and safety logs must be restricted to authorized guardians via secure authentication.

- **Reliability (Offline Resilience):** The application must include a local SQLite buffering mechanism to log safety events during internet outages and synchronize them immediately upon reconnection.
- **Usability:** The guardian dashboard must feature high-contrast UI elements and accessible fonts for quick information retrieval during high-stress emergency scenarios.
 - The user interface for the VIB individual must be entirely voice-responsive or accessible via high-tactile physical buttons.
- **Availability:** The cloud-synchronized safety monitoring service should maintain an uptime of 99.9% to ensure constant guardian connectivity.

5.5. System Requirements (Technical Environment)

- **Software:** Flutter or React Native for the mobile app, Firebase for the cloud backend, and Python for the edge-processing scripts on Raspberry Pi.
- **Hardware Interfacing:** Bluetooth Low Energy (BLE) or Wi-Fi for communication between the mobile app and the Raspberry Pi unit.
 - Data Capture - Build a web app using HTML5 Canvas in JavaScript to record x, y coordinates, time, pressure, stroke ID.
 - Store raw and processed data securely with encryption.
 - Use participant IDs, without personal information and maintain consent logs.

5.6. Use Case Analysis

The following use case describes the primary interaction within your component.

Use Case ID	UC-01
Use Case Name	Emergency Alert Escalation
Actors	Visually Impaired User, Smart Spectacle, Mobile App, Remote Guardian
Pre-conditions	Spectacles are powered on and connected to the Guardian's mobile app.
Main Flow	1. Spectacle sensors detect a fall or 60 seconds of inactivity. 2. App initiates a voice verification check on the user's phone. 3. User does not cancel the alert within 10 seconds. 4. App sends a high-priority SOS with indoor location to the Guardian.
Post-conditions	Guardian receives the alert and environmental context to initiate rescue.

Table 1: Use Case

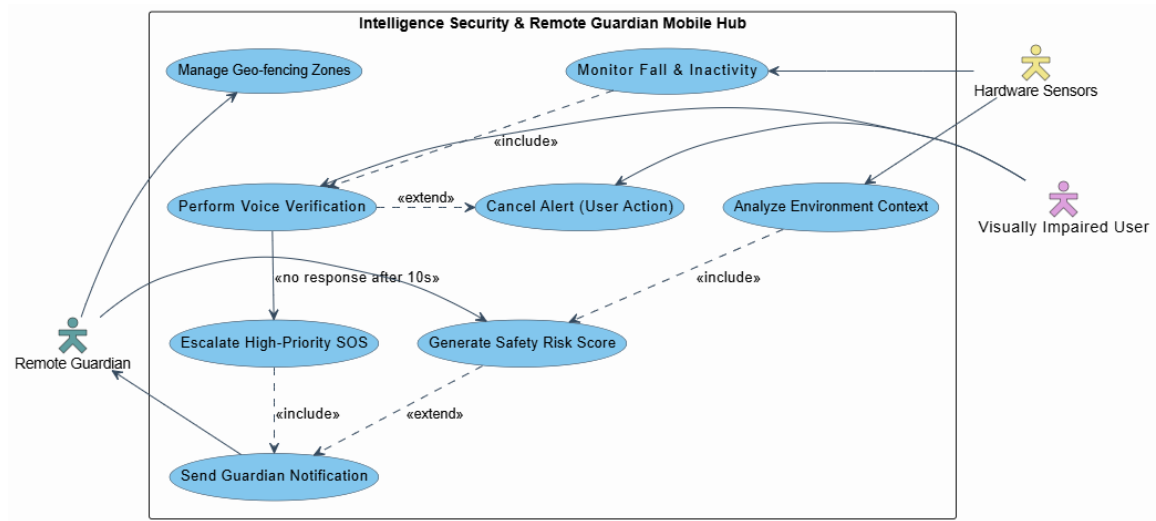


Figure 4: Use case Diagram.

6. Commercialization Plan

6.1. Target Market and Customer Persona

The primary target market for the proposed smart spectacle system focuses on visually impaired individuals requiring autonomous navigation assistance in indoor settings such as homes, offices, and educational institutions.

- **Target Market Segments:**

- Visually impaired individuals seeking affordable assistive navigation technology.
- Rehabilitation centers and specialized hospitals for the blind.
- Educational institutions support students with visual disabilities.
- Assistive technology organizations and accessibility programs.

- **Customer Persona:** A visually impaired person in need of a lightweight, wearable, and affordable assistive device that facilitates safe movement and navigation.

6.2. Value Proposition

The **Intelligence Security & Remote Guardian System** offers a unique "Peace of Mind" value proposition by bridging the communication gap between the visually impaired user and their caregiver through proactive intelligence.

User / Guardian Problem	Proposed Solution (Your Component)	Value Proposition
User: Fear of being helpless during a fall or sudden medical emergency when alone.	Automated Inactivity & Fall Monitoring: Uses IMU sensor fusion to detect emergencies without user intervention.	Safety Net: Provides independence to the user, knowing that help will be summoned automatically even if they are unconscious.
User: Difficulty in manually operating a phone to call for help during distress.	Hands-free Voice-Activated SOS: Integrated NLP engine that triggers alerts via specific voice commands.	Accessibility: Ensures a 100% hands-free emergency trigger mechanism, essential for physically impaired or panicked users.
Guardian: Stress due to lack of specific information (only GPS) during an alert.	Context-Aware Alerting: Displays specific environmental hazards (e.g., "User near stairs") on the dashboard.	Proactive Intervention: Moves beyond raw coordinates to provide "meaningful alerts," allowing the guardian to understand the exact nature of the risk.
Guardian: High rate of false alarms from basic motion sensors causing "notification fatigue."	Multi-stage Verification Logic: A 10-second voice verification window for the user to cancel accidental alerts.	Reliability: Ensures the guardian is only notified of genuine emergencies, maintaining the system's credibility and trust.
System: Loss of safety monitoring when internet connectivity is unstable.	Hybrid Local/Cloud Sync: Local SQLite buffering and edge-based logic for	Resilience: Guarantees that the "Guardian Hub" remains functional and logs

	immediate user-side feedback.	data even during temporary network dropouts.
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Table 2: Value Proposition

6.3. Revenue Model

The project will utilize a product-based revenue model, selling the device directly to customers or through assistive technology distributors. Potential revenue sources include:

- Direct sales of the smart spectacle device.
- Partnerships with rehabilitation centers and hospitals.
- Government or accessibility program support.
- Licensing the technology to established assistive device manufacturers.

6.4. Cost Estimation

Component	Estimated Cost (LKR)
Raspberry Pi Zero 2 W	8,890
Camera Module V2	8,000
LiDAR Sensor	1,550
IMU Sensor	680
Power System	1,700
Other Components	5,000
Estimated Prototype Cost	~ 25,820 LKR

Table 3: Cost Estimation

6.5. Pricing Strategy

The product pricing strategy will focus on affordability and accessibility.

Possible pricing approaches include:

- Affordable pricing model targeting developing countries
- Subsidized pricing through government or healthcare programs
- Partnering with NGOs that support visually impaired individuals

The retail price of the device will be around 30,000 - 35,000 LKR.

6.6. Competitive Advantage

- **Improved Obstacle Detection:** Integration of multiple sensors (IMU, LiDAR, Camera) provides superior awareness compared to single-sensor devices.
- **Edge-based Processing:** Critical safety features, such as fall detection and obstacle recognition, operate without needing internet connectivity.
- **Context-Aware Guardian Hub:** Unlike existing apps, this system provides guardians with environmental context (e.g., "User near stairs") rather than just GPS coordinates.
- **Affordability:** significantly lower cost compared to high-end commercial smart glasses.

6.7. Intellectual Property Considerations

The proposed system includes novel elements related to multi-sensor integration for assistive navigation.

- **Patent Protection:** Seeking patents for the hardware layout, sensor integration methods, and system design.
- **Copyright Protection:** Protecting the source code, software algorithms, and custom AI models used in the mobile hub and edge tier.
- **Collaboration:** Exploring opportunities with technology organizations for commercialization and IP management.

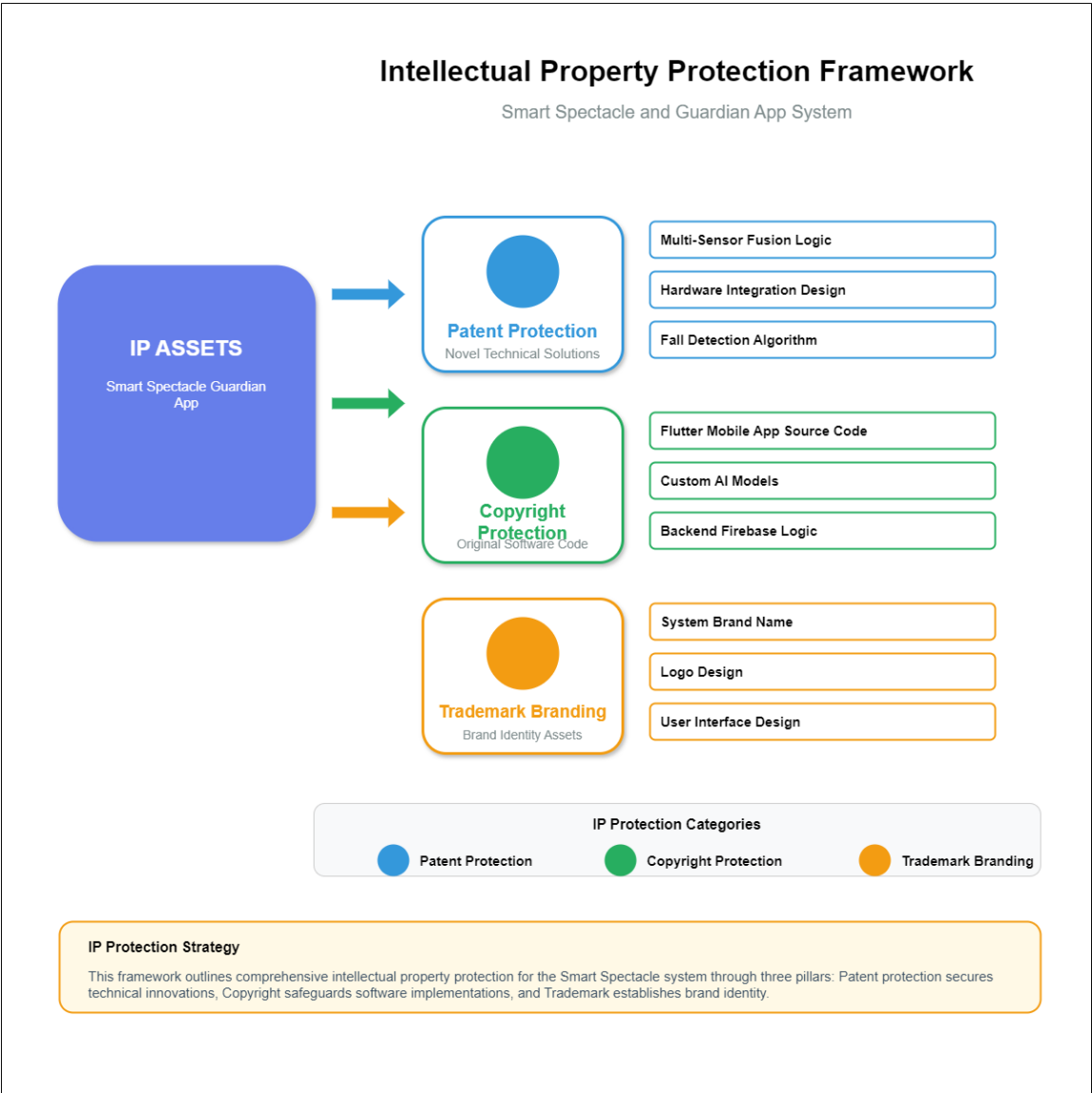


Figure 5:IP Protection Framework

7. Budget And Justification

7.1. Estimated Budget Breakdown

Component	Quantity	Estimated Cost (LKR)
Raspberry Pi Zero 2 W	1	8,890
Raspberry Pi Camera Module V2	1	7,950
VL53L1X LiDAR Sensor	1	1,550
MPU6050 IMU Sensor	1	680
Li-Po Battery (3.7V 3000mAh)	1	1,550
MT3608 Boost Converter	1	150
MicroSD Card (32GB)	1	3,100
Earphone	1	810
Sunglass Frame and Wiring	1	1,000
Estimated Prototype Cost		~ 25,680 LKR

Table 4: Estimated Budget Breakdown for Prototype Development

7.2. Expense Justification

7.2.1. Hardware Costs

Most of the budget is allocated to hardware components essential for the real-time processing and safety features of the system:

- **Raspberry Pi Zero 2 W:** Serves as the central embedded processing unit for object detection and alert logic.
- **Camera Module V2:** Facilitates data collection for obstacle recognition.
- **VL53L1X LiDAR Sensor:** Used for precise distance detection, specifically for identifying hazards like staircases.

- **MPU6050 IMU Sensor:** Critical for the Intelligence Security component to detect head orientation and sudden falls, improving navigation accuracy.
- **Li-Po Battery & Boost Converter:** Ensures the system is portable and provides a stable power supply to the Raspberry Pi.
- **MicroSD Card:** Stores the operating system, custom software programs, and machine learning models.
- **Earphone:** Provides essential audio feedback (voice triggers/alerts) to the user.
- **Sunglass Frame & Wiring:** This will be used for the connection of all the hardware components to the object detection system.

7.2.2. Software Requirements and Licenses

The proposed system and the Guardian Mobile Hub primarily utilize open-source software tools and frameworks, which significantly minimize development and deployment costs. The selected software stack is justified by its high performance in real-time safety monitoring.

Software Tool / Framework	Purpose	License Cost
Python	Hardware-level sensor data processing and Fall Detection logic.	Free
Flutter SDK	Developing the cross-platform Guardian Mobile Application (Android/iOS).	Free
Firebase Services	Real-time database, Cloud Messaging (alerts), and User Authentication.	Free Tier
OpenCV / TF Lite	Vision-based hazard detection and AI model optimization.	Free

VS Code / Android Studio	Primary Integrated Development Environments (IDEs) for coding.	Free
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Table 5: Software Requirements and License

7.3. Cloud Services

The system also employs an edge computing architecture, which means that most of the data processing is done on the hardware itself. This is not a decision that is based on expensive cloud computing solutions, and there are no costs involved in cloud services in this prototype.

7.4. Human Effort Cost

The members of the research team are working on the development as a part of their academic research project. Hence, there is no direct financial cost involved in human resources during the phase of developing the prototype.

8. Work Breakdown Structure (WBS)

8.1. Task Breakdown per Member

Member	System Component	Key Tasks
Member 1	Intelligent Vision-Based Danger Detector 	Task 1: Collect and prepare image dataset for indoor hazards. Task 2: Configure YOLO object detection model for edge devices. Task 3: Train and optimize the model using labeled datasets. Task 4: Evaluate detection accuracy and performance. Task 5: Integrate detection system with the smart spectacle platform.
Member 2	Intelligent Security & Remote Guardian System 	Task 1: Design and develop the guardian mobile application. Task 2: Implement fall detection and inactivity monitoring algorithms. Task 3: Develop geo-fencing functionality for safe zones. Task 4: Integrate cloud database for real-time monitoring.

		Task 5: Implement SOS emergency alert system.
Member 3 (ME)	Smart Spectacle Hardware Module with Multi-Sensory Integration 	Task 1: Design the wearable smart spectacle hardware structure. Task 2: Integrate Raspberry Pi with camera, LiDAR, and IMU sensors. Task 3: Implement sensor data acquisition and synchronization. Task 4: Develop audio output interface for navigation alerts. Task 5: Implement push-button control for system interaction.
Member 4	Hybrid Indoor Positioning System with Offline Capability 	Task 1: Develop indoor localization algorithm using IMU sensors. Task 2: Implement vision-based landmark recognition. Task 3: Integrate dead-reckoning for indoor positioning. Task 4: Combine vision and IMU data for hybrid localization. Task 5: Integrate positioning system with navigation and hazard detection modules.

Table 6: Task Breakdown per Member for the Proposed Smart Spectacle System

8.2. Timeline Alignment

The development process of the proposed IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired will be carried out in an aligned manner with an appropriate timeline so that it can be achieved systematically during the entire research period. The project will be achieved through different key milestones that will be included in the research process.

The project will be aligned with the following timeline:

- Phase 1 – Literature Review and Problem Analysis (Month 1):

Thorough research will be performed based on the existing research works on assistive technologies, indoor navigation systems, computer vision models, and IoT-based wearable devices.

- Phase 2 – System Architecture and Design (Month 2):

Design of the entire system, including hardware as well as software components, shall be conducted.

- Phase 3 – Hardware and Software Development (Month 3-4):

The development of the individual components of the entire system will be performed.

- Phase 4 – System Integration and Testing (Month 5):

All the components of the entire system will be integrated and the testing of the entire system will be performed.

- Phase 5 – Evaluation and Documentation (Month 6):

Analysis of the entire system, completion of the research report, and completion of the project's presentation.

8.3. Balanced Workload Distribution

The project workload is distributed among the four members of the group based on the technical expertise of each member and the needs of the proposed project. Each member of the group is responsible for the development of a particular module of the project, including vision-based hazard detection, the remote guardian monitoring system, the smart spectacle hardware module, and the hybrid indoor positioning system. This ensures the development of all the major parts of the project efficiently, with all the members of the group participating equally.

Besides the development of the project's individual parts, some activities such as integration, testing, and report writing will be undertaken jointly by all the members of the

group to ensure the equal participation of all members in the successful completion of the research project.

9. Gantt Chart



Figure 6: Gantt Chart for the Proposed Smart Spectacle Research Project

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11. Appendices

11.1. Appendix A - Survey Instruments

As a follow-up, the following is a sample questionnaire that seeks to understand the challenges faced by visually impaired individuals when moving around within an indoor environment. Additionally, the questionnaire seeks to find out the possible application of wearable assistive technology devices.

Target Audience: The target audience for the survey will be individuals who are visually impaired, caregivers of individuals who are visually impaired, and professionals who are knowledgeable about assistive technologies.

Survey Questions

1. What type of difficulties do visually impaired individuals commonly face when navigating indoor environments?
 - Difficulty identifying obstacles.
 - Difficulty recognizing stairs or elevation changes.
 - Difficulty determining direction or location.
 - All the above
2. Which assistive device is commonly used by visually impaired individuals for mobility?
 - White cane
 - Guide dog
 - Smartphone navigation applications
 - Wearable assistive devices
3. How important is obstacle detection for safe indoor navigation?
 - Particularly important
 - Important
 - Moderately important

- Not important
4. What type of feedback would be most useful for a smart navigation system?
- Audio instructions through earphones
 - Vibration alerts
 - Visual notifications
 - Combination of audio and vibration
5. How useful would a wearable smart spectacle device be for assisting visually impaired individuals in indoor navigation?
- Particularly useful
 - Useful
 - Slightly useful
 - Not useful
6. Which feature would be most important in a smart navigation device for visually impaired users?
- Obstacle detection
 - Staircase detection
 - Real-time navigation assistance
 - Emergency alert system

11.2. **Appendix B - Detailed System Diagrams**

11.2.1. **Smart spectacle hardware architecture diagram**

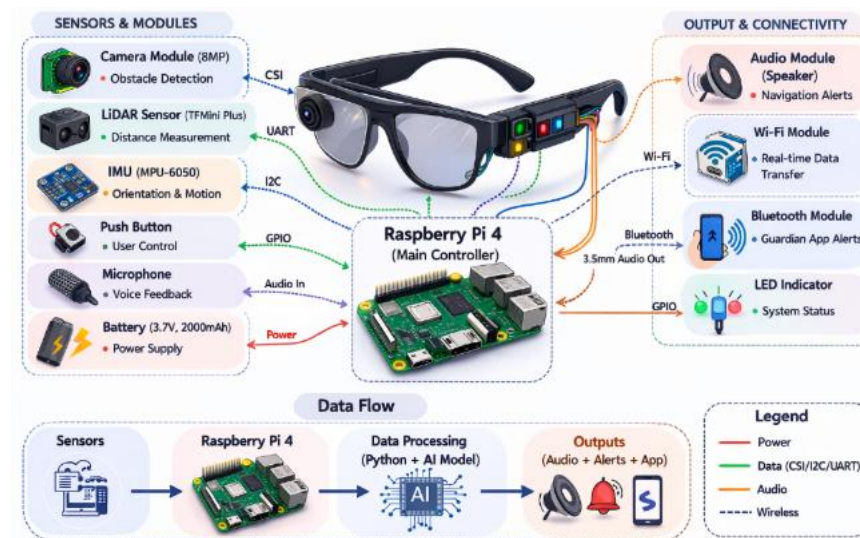


Figure 7: Smart Spectacle Hardware Architecture Diagram

11.2.2. Data processing pipeline

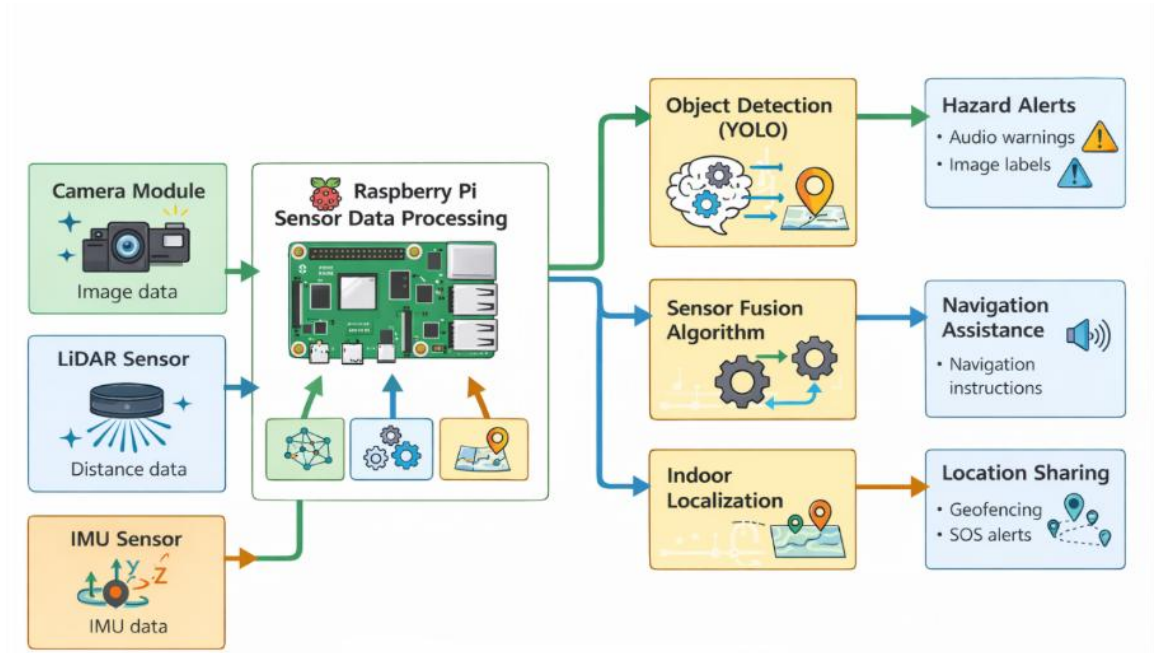


Figure 8: Sensor Data Processing Pipeline

11.3. Appendix C – Preliminary UI Mockups

This appendix presents the preliminary user interface designs for the Guardian Monitoring Mobile Application, which allows guardians to monitor the safety and location of visually impaired users using the proposed smart spectacle system.

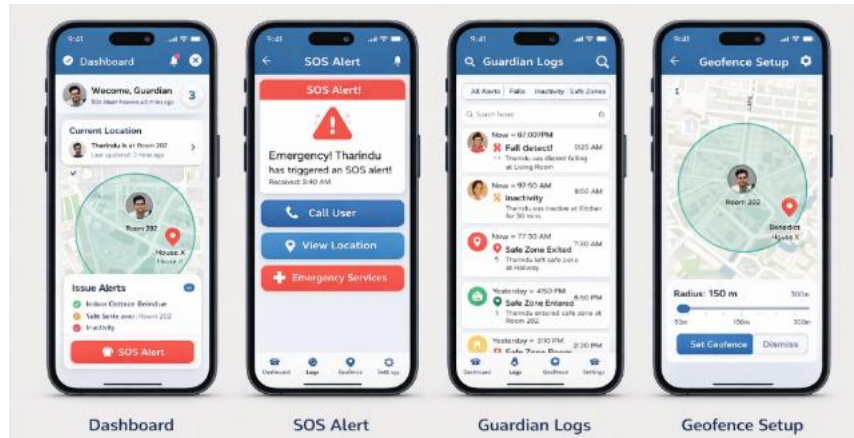


Figure 9: Preliminary User Interface Mockups for the Guardian Monitoring Mobile Application

11.4. Appendix D – Risk Analysis

Risk Factor	Impact	Probability	Mitigation Strategy
Network Latency: Delay in syncing SOS alerts from the spectacle to the Guardian App via Firebase.	Delays the "Golden Hour" medical response for the user.	Medium	Utilize Firestore with high-priority headers and implement MQTT for real-time heartbeat monitoring.
False Positives: Fall detection triggered by non-emergency activities (e.g., sitting down abruptly).	Causes "Alert Fatigue" for the guardian, reducing trust.	High	Implementation of the 10-second Voice Verification Window (UC-01) allowing the user to cancel accidental alerts locally.
Connectivity Loss: The Guardian Hub losing internet	Leaves the user completely vulnerable without a safety net.	Medium	To address potential health emergencies where the user may be incapacitated, the system prioritizes

connection while the user is navigating.			Automated Behavioral Analysis over simple connectivity checks. By monitoring for extended periods of inactivity and utilizing keyword-based voice recognition (e.g., SOS triggers from the user or bystanders), the system ensures the Guardian is alerted even when the user cannot manually interact with the device.
Data Privacy Breach: Unauthorized access to the user's location or environmental risk data.	Potential legal issues and loss of user privacy/trust.	Low	Enforce Firestore Security Rules and End-to-End Encryption (E2EE) for all risk metadata synced between the spectacle and the mobile app.
Battery Optimization: Background execution of the Guardian App causing excessive battery drain.	Leads to system downtime during critical navigation periods.	Medium	Optimize Flutter background services using Isolates and adjust data polling frequency based on the user's "Safety Zone" status.
Notification Fatigue: The caregiver ignoring alerts due to	Degrades the guardian's user experience and app engagement.	Medium	Implement Tiered Alerting: Minor hazards are silent updates, while Falls/SOS triggers an

frequent minor risk updates.			"Override Do Not Disturb" (DND) audio alarm.
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Table 7: Risk Analysis for the Smart Spectacle Hardware Module

11.5. Appendix E – Ethical Considerations

This research is carried out following the guidelines on research ethics provided by the Sri Lanka Institute of Information Technology. This institute has an Ethics Review Committee (ERC), which deals with the review of the research proposal and ensures that all the research activities are carried out ethically.

The following ethical principles are considered in this research project:

- **Personal Data:**

This research does not require any personal data from any individual. If any data is required in the experiment for the system, it is related to objects in the environment, such as furniture, doors, staircase, etc.

- **Privacy protection:**

If images are captured during experiments in indoor environments, they will not intentionally include identifiable individuals. Any accidental appearance of people will not be stored or used for analysis.

- **Responsible use of technology:**

The developed smart spectacle system is only intended to assist and enhance the safety and mobility of people with visual impairments.

- **Compliance with institutional ethics guidelines:**

This research has been conducted based on the ethical guidelines suggested by the SLIIT ERC to carry out ethical academic research.

- **Academic integrity:**

All the information sources, data sets, and research material used in the current study are properly credited to ensure academic integrity.

SLIIT Ethics Review Committee: <https://www.sliit.lk/about/about-sliit/ethics-review-committee>